torch.linalg.vector_norm#

https://pytorch.org/docs/stable/generated/torch.linalg.vector_norm.html

Description:

Computes a vector norm. If x is complex valued, it computes the norm of x.abs() Supports input of float, double, cfloat and cdouble dtypes. This function does not necessarily treat multidimensional x as a batch of vectors, instead: If dim= None, x will be flattened before the norm is computed.

Function Signature

```
torch.linalg.vector_norm(x, ord=2, dim=None, keepdim=False,
*, dtype=None, out=None) → Tensor#

Parameters

Keyword Arguments

Returns
```

Parameters

Keyword

out (Tensor, optional) – output tensor. Ignored if None. Default: None. dtype (torch.dtype, optional) – type used to perform the accumulation and the return. If specified, x is cast to dtype before performing the operation, and the returned tensor's type will be dtype if real and of its real counterpart if complex. dtype may be complex if x is complex, otherwise it must be real. x should be convertible without narrowing to dtype. Default: None

Returns:

A real-valued tensor, even when x is complex.

Code Examples

```
>>> from torch import linalg as LA
>>> a = torch.arange(9, dtype=torch.float) - 4
>>> a
tensor([-4., -3., -2., -1.,  0.,  1.,  2.,  3.,  4.])
>>> B = a.reshape((3, 3))
>>> B
tensor([[-4., -3., -2.],
        [-1.,  0.,  1.],
        [ 2.,  3.,  4.]])
>>> LA.vector_norm(a, ord=3.5)
tensor(5.4345)
>>> LA.vector_norm(B, ord=3.5)
tensor(5.4345)
```

Notes & Warnings

> **NOTE** See also torch.linalg.matrix_norm() computes a matrix norm.

Detailed Documentation

Documentation

Computes a vector norm.

If x is complex valued, it computes the norm of x.abs()

Supports input of float, double, cfloat and cdouble dtypes.

This function does not necessarily treat multidimensional x as a batch of vectors, instead:

If dim= None, x will be flattened before the norm is computed.

If dim is an int or a tuple, the norm will be computed over these dimensions and the other dimensions will be treated as batch dimensions.

This behavior is for consistency with torch.linalg.norm().

ord defines the vector norm that is computed. The following norms are supported:

sum(abs(x)^{ord})^{(1 / ord)}

where inf refers to float('inf'), NumPy's inf object, or any equivalent object.

dtype may be used to perform the computation in a more precise dtype. It is semantically equivalent to calling linalg.vector_norm(x.to(dtype)) but it is faster in some cases.

torch.linalg.matrix_norm() computes a matrix norm.

x (Tensor) – tensor, flattened by default, but this behavior can be controlled using dim. (Note: the keyword argument input can also be used as an alias for x.)

ord (int, float, inf, -inf, 'fro', 'nuc', optional) – order of norm. Default: 2

dim (int, Tuple[int], optional) – dimensions over which to compute the norm. See above for the behavior when dim= None. Default: None

keepdim (bool, optional) – If set to True, the reduced dimensions are retained in the result as dimensions with size one. Default: False

out (Tensor, optional) – output tensor. Ignored if None. Default: None.

dtype (torch.dtype, optional) – type used to perform the accumulation and the return. If specified, x is cast to dtype before performing the operation, and the returned tensor's type will be dtype if real and of its real counterpart if complex. dtype may be complex if x is complex, otherwise it must be real. x should be convertible without narrowing to dtype. Default: None

A real-valued tensor, even when x is complex.
